# COURSE OUTLINE

| | |
|---|---|
| **PROGRAMME TITLE** | **Advanced Machine Learning in Production: From Leakage-Free Pipelines to Deployed Models** |
| **OVERVIEW** (brief introduction of the topic) | Most organisations now use AI, yet far fewer can take a model from a notebook to a service that stays accurate, explainable and monitored in production. This programme closes that gap for practitioners who already build models.<br><br>**Adoption is high, but production is the bottleneck**<br>• According to the Stanford AI Index (Maslej et al., 2025), 78% of organisations reported using AI in 2024, up from 55% the year before.<br>• A Gartner survey of 644 organisations in the United States, Germany and the United Kingdom found that fewer than half of AI projects reach production, and that moving from prototype to production takes about eight months on average (Gartner, as cited in Wilkinson, 2024).<br>• Sculley et al. (2015) showed that real-world machine learning systems commonly incur large ongoing maintenance costs, which they describe as hidden technical debt.<br><br>**Models fail quietly when the data pipeline is wrong**<br>• Kapoor and Narayanan (2023) found data leakage in 17 scientific fields, collectively affecting 294 published papers, in some cases producing overoptimistic conclusions.<br>• The scikit-learn documentation warns that leakage gives overly optimistic performance estimates and poorer performance in production, and recommends splitting the data before any preprocessing and using a pipeline (scikit-learn developers, n.d.).<br><br>**The skills gap is the main constraint**<br>• The World Economic Forum (2025) reports that 63% of employers identify skill gaps as the biggest barrier to business transformation, with AI and big data ranked as the fastest-growing skills.<br>• In Malaysia, 59% of organisations have deployed or are piloting AI, yet only 5% are confident they can recruit and retain enough AI-skilled talent (Aon, 2026).<br><br>**Why this toolset**<br>• The programme uses open-source tools that participants can adopt at no licence cost: scikit-learn, Optuna, MLflow, SHAP, FastAPI and Docker.<br>• The same workflow carries over to Azure Machine Learning, a cloud service for training and deploying models and managing MLOps, with MLflow integration (Microsoft, 2026). |

| | |
|---|---|
| | This hands-on programme takes one running case study, a loan-default model, through the production lifecycle in four hours. Participants build a leakage-free pipeline, run it as a modular training script, tune and track experiments, explain a prediction to a business audience, check for drift, and serve the model through a containerised API. The trainer then demonstrates the same workflow on Azure Machine Learning. |
| **LEARNING OUTCOME** | By the end of the programme, participants will be able to:<br>• Select the appropriate approach (classic machine learning, deep learning or a large language model) for a given business problem<br>• Detect data leakage and build a leakage-free scikit-learn pipeline with a validation strategy that fits the data<br>• Organise training code into modular, configurable and reproducible scripts<br>• Tune hyperparameters with Optuna and track experiments with MLflow<br>• Explain model predictions with SHAP and translate them into business language<br>• Detect data drift using the Population Stability Index (PSI) and the Kolmogorov-Smirnov (KS) test, and set alert thresholds<br>• Serve a trained model as a REST API with FastAPI inside a Docker container<br>• Describe what to monitor in production and how the same workflow runs on Azure Machine Learning |
| **DURATION** (days) | 0.5 day (4 hours, 240 minutes) |
| **TARGET PARTICIPANTS** (position/level) | • Data scientists and machine learning engineers moving models into production<br>• Data analysts and data engineers extending into MLOps<br>• Software engineers who integrate or operate machine learning services<br>• Technical leads and analytics managers responsible for model quality and governance<br><br>**Prerequisite:** working knowledge of Python, pandas and scikit-learn, including training and evaluating a supervised model. Participants need a laptop with Python 3.11, Git and Docker Desktop installed, and the course repository cloned from the students branch before the session. No Azure account is required; the cloud segment is a trainer demonstration. |

| | |
|---|---|
| **COURSE CONTENT** (with submodules) | **Module 1: Trustworthy Data and Leakage-Free Pipelines (55 min)**<br>• **Submodule 1: Choosing the right approach** – where classic machine learning, deep learning and large language models fit, mapped to real business problems<br>• **Submodule 2: Data quality and data leakage** – the main data quality dimensions, and target, train-test, temporal and group leakage<br>• **Submodule 3: Finding leakage** – hands-on search for every leakage source in a loan-default dataset<br>• **Submodule 4: Leakage-free pipelines and validation** – the scikit-learn Pipeline, splitting before preprocessing, and time-ordered validation<br><br>**Module 2: Production Training, Tuning and Tracking (55 min)**<br>• **Submodule 1: From notebook to modular scripts** – single-responsibility modules, YAML configuration, seeds and logging, and running the training entry point from the command line<br>• **Submodule 2: Hyperparameter optimisation** – grid, random and Bayesian search (Bergstra & Bengio, 2012), with hands-on tuning in Optuna<br>• **Submodule 3: Experiment tracking** – logging parameters, metrics and artefacts with MLflow, and comparing runs<br><br>**Module 3: Explainability and Drift Monitoring (55 min)**<br>• **Submodule 1: Explaining models with SHAP** – global summary plots and local waterfall plots (Lundberg & Lee, 2017)<br>• **Submodule 2: From explanation to business narrative** – writing a plain-language explanation of one prediction for a non-technical stakeholder<br>• **Submodule 3: Detecting data drift** – feature, concept and label drift, measured with PSI and the KS test<br>• **Submodule 4: Anomaly detection and alert thresholds** – trainer demonstration of Isolation Forest, then setting thresholds and a monitoring cadence<br><br>**Module 4: Serving, Monitoring and Cloud (60 min)**<br>• **Submodule 1: Serving a model as a containerised API** – FastAPI with Pydantic validation, built and run in Docker and tested with a curl request<br>• **Submodule 2: The MLOps dashboard** – model performance, data health, system health and business impact, read from a six-panel dashboard<br>• **Submodule 3: Moving to Azure Machine Learning** – trainer demonstration of the workspace, a cloud training job and the model registry<br>• **Submodule 4: Back-to-work action plan and wrap-up** – personal next steps, questions, key takeaways and feedback |

| | |
|---|---|
| **LEARNING METHODOLOGIES** | Interactive and experiential learning with short lectures, live coding demonstrations, guided hands-on exercises in Jupyter notebooks, group discussion, a running loan-default case study and a checkpoint at the end of every module. |

References


Aon. (2026, August 19). *Organisations in Malaysia face increasing AI talent gap and workforce readiness challenges, Aon study finds* [Press release]. https://www.aon.com/apac/in-the-press/asia-newsroom/2026/organisations-in-malaysia-face-increasing-ai-talent-gapand-workforce-readiness-challenges-aon-study-finds

Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research, 13*(10), 281–305. https://www.jmlr.org/papers/v13/bergstra12a.html

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns, 4*(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*. https://arxiv.org/abs/1705.07874

Maslej, N., et al. (2025). *The AI Index 2025 annual report*. Institute for Human-Centered AI, Stanford University. https://hai.stanford.edu/ai-index/2025-ai-index-report

Microsoft. (2026, September 4). *What is Azure Machine Learning?* Microsoft Learn. https://learn.microsoft.com/en-us/azure/machine-learning/overview-what-is-azure-machine-learning

scikit-learn developers. (n.d.). *Common pitfalls and recommended practices*. scikit-learn documentation. https://scikit-learn.org/stable/common_pitfalls.html

Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). Hidden technical debt in machine learning systems. *Advances in Neural Information Processing Systems, 28*, 2503–2511. https://research.google/pubs/hidden-technical-debt-in-machine-learning-systems/

Wilkinson, L. (2024, May 17). *Enterprises struggle to show the value of AI projects*. CIO Dive. https://www.ciodive.com/news/generative-ai-adoption-barrier-project-value/716504/

World Economic Forum. (2025, January 8). *Future of Jobs Report 2025: 78 million new job opportunities by 2030 but urgent upskilling needed to prepare workforces* [Press release]. https://www.weforum.org/press/2025/01/future-of-jobs-report-2025-78-million-new-job-opportunities-by-2030-but-urgent-upskilling-needed-to-prepare-workforces/